LEARNER GUIDE

Showing solutions on a number line

Inequalities · Lesson 3. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

An inequality such as $x \geq 2$ has far too many answers to list, but a **number line** can show them all at once. A **circle** marks each boundary: **filled** when the boundary is included (≤ or ≥), **open** when it is not (< or >). An **arrow** shows that the set carries on for ever; a **segment** between two circles shows a set trapped between two boundaries. The same set can also be written in **words**, in **symbols**, or as a **list** of whole numbers.

What you are learning to notice

- Filled circle: the boundary is included (≤ or ≥). Open circle: it is not (< or >).
- One boundary: one circle and an arrow. Two boundaries: two circles and the segment between them.
- To solve a double-ended inequality, do the same to **all three parts**.
- Two rules together allow only the numbers where their pictures **overlap** — and sometimes there is no overlap.

Moving through it, one step at a time

Find the boundary first and mark it with a circle. Ask: is the boundary itself included? If yes, fill the circle. Then decide which way the set goes, and draw the arrow or the segment. To check, read your picture back in words, and test one number inside and one outside.

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — Drawing a one-sided inequality

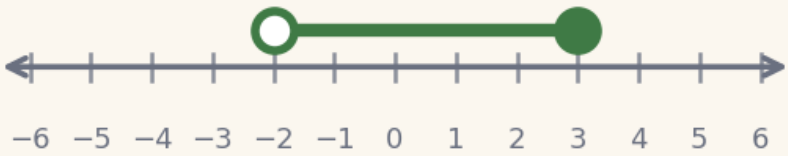
Draw $x \geq -1$.

1. The boundary is –1. Is –1 included? ≥ means yes.	filled circle at –1
2. x is greater than –1, so the set goes to the right.	arrow to the right
3. Whole numbers in the set:	–1, 0, 1, 2, …

A filled circle at –1 with an arrow to the right.

Example 2 — Solving and drawing a double-ended inequality

Solve $-1 < 2x + 3 \leq 9$ and draw the answer.



1. Subtract 3 from all three parts.	$-4 < 2x \leq 6$
2. Divide all three parts by 2.	$-2 < x \leq 3$
3. Open circle at –2 (not included), filled at 3 (included).	whole numbers –1, 0, 1, 2, 3

$-2 < x \leq 3$: a segment from an open circle at –2 to a filled circle at 3.

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (placing whole numbers, including negatives, on a number line, and solving inequalities from Lesson 2), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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